Aigens: Delivering a powerful ordering and delivery platform for hotels and restaurants at scale

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About Aigens

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Founded in 2012, Aigens offers end-to-end solutions to major restaurants, serving in-store POS and CRM needs as well as integration with payment and online delivery platforms. The company operates across 12 countries and serves over 4,000 restaurant locations including major local and global brands.

Industries: Retail & Consumer Goods

Restaurant and h
platform Aigens l
Google Cloud nat
day one, finding t
platform's serverl
architecture to be
home for iterative

and rapid deploy
clients.

Google Cloud results

- Terabytes of data in BigQuery hosted at minimal cost under pay per use model
- Serverless infrastructure has saved Aigens thousands of engineering hours
- DevOps team can use canary deployments and switch versions or rollback in less than 10 seconds

Deplo
enter
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Location: Hong Kong

Cloud Run (https://cloud.google.com/run/),

VPN

Firebase

Looker

tudio)

Vertex AI (https://cloud.google.com/vertex-ai/)

Google Cloud (https://cloud.google.com/)

BigQuery (https://cloud.google.com/bigquery/)

Cloud Bigtable (https://cloud.google.com/bigtable/)

Cloud CDN (https://cloud.google.com/cdn/)

Cloud Load Balancing (https://cloud.google.com/load-balancing/)

Logging (https://cloud.google.com/logging/)

Cloud Run (https://cloud.google.com/run/)

Cloud SQL (https://cloud.google.com/sql/)

Compute Engine (https://cloud.google.com/compute/)

Cloud VPN (https://cloud.google.com/hybrid-connectivity/)

Cloud Firestore (https://cloud.google.com/firestore/)

Firebase Realtime Database (https://firebase.google.com/products/realtime-database/)

App Engine (https://cloud.google.com/appengine/)

Looker Studio (https://cloud.google.com/looker-studio)

Cloud Memorystore (https://cloud.google.com/memorystore/)

- **Centralized logging in Google Cloud enables issues to be identified within minutes**

Cloud SQL for PostgreSQL (https://cloud.google.com/sql/docs/postgres/)

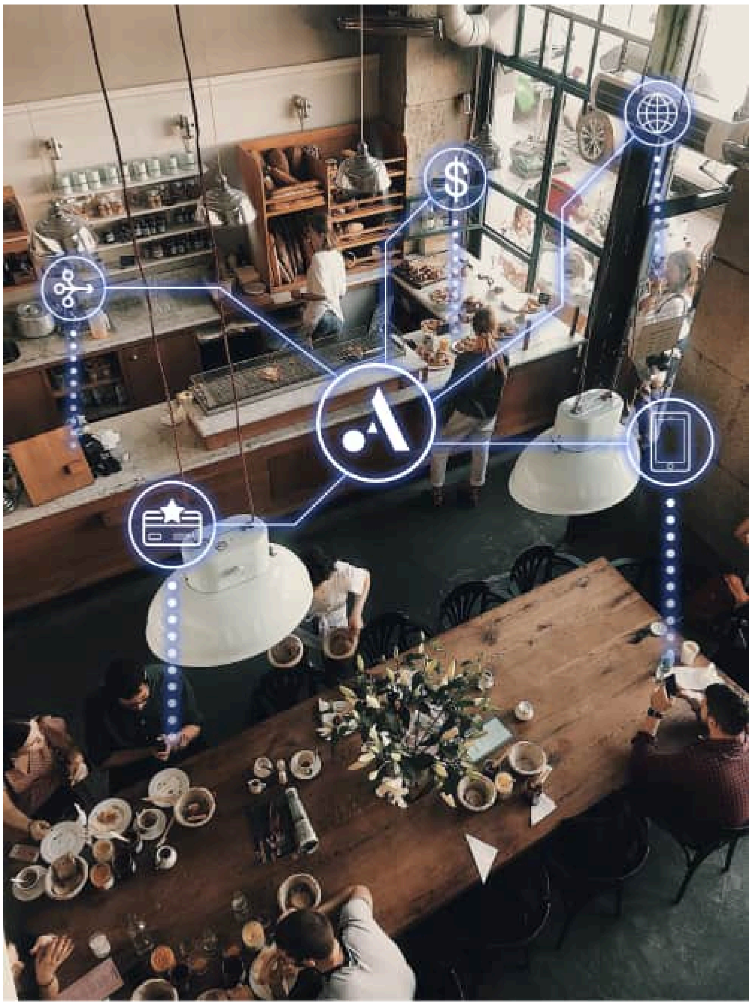
Vertex AI (https://cloud.google.com/vertex-ai/)

It's only just been a decade since the restaurant industry began its technology revolution. Built on a long tradition of deeply physical dining experience, it took a first wave of entrepreneurs to reveal the possibilities of smarter and simpler ordering for customers, wherever they may be.

For Aigens (https://www.aigens.com/) co-founder Peter Liu, it was returning to Hong Kong after his studies at UC Berkeley that made him wish there was a way to order dinner with the technology that excited him.

"Our original mission was to enable restaurants to use technology to streamline operations," says Peter. "We concepted using our phones to scan QR codes and use NFC to access menus in web formats. But back in 2012 there was little knowledge around that, so we moved to more concrete ideas like self-serve kiosks as our starting place until the world caught up on smartphones."

From kiosks in Hong Kong and into online ordering across 12 countries, Aigens has connected the physical world of restaurants with the online world that became so crucial to the food and beverage industry during the pandemic. Aigens made it easier for its customers to list their menus across Asia's many food delivery services and super apps, as well as accept payments across varied regional payment platforms.



Fully managed scalability provides the ideal baseline

Today, Aigens offers a robust central platform that opens up an ecosystem approach for its client base of enterprise-level restaurant and hospitality chains. With more than 20 million users flowing through its platform to over 4,000 restaurant locations, Peter felt that the right platform for Aigens had to be scalable to meet the fluctuating needs of the food serving industry as well as their own growth trajectory. He found that their choice to build on Google App Engine (https://cloud.google.com/appengine) at the very beginning put Aigens into just the right environment to build and grow with confidence.

In Aigens comparison of platforms, Peter says they found Google Cloud (https://cloud.google.com/) to be easy to use and has all the different technologies and solutions they have needed as they have grown. He puts the time saved by working on a fully managed platform in "thousands of hours of engineering time."

"With an entirely serverless environment, I don't have to worry about scaling," says Peter. "Add a fully managed data store and I don't have to worry about backups, encryption, or anything else. It means our developers can code the application instead of setting up infrastructure. We'd much rather spend our time developing business logic than managing infrastructure."

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Enterprise-grade security embedded in the system

Working with some of the biggest global brands in fast service restaurants, Aigens prides itself in providing robust security measures to ensure the protection of sensitive data and transactions for its clients, many of which are leading global brands in the fast service restaurant industry. "Enterprise clients often inquire about the scalability and security of our solutions, and we are proud to highlight the Google Cloud credentials and certifications that provide peace of mind for these critical components," says Peter.

In some cases, enterprise customers may require a VPN to connect the cloud platform to their local systems. To accommodate this requirement, Aigens uses Cloud VPN (https://cloud.google.com/network-connectivity/docs/vpn/concepts/overview) and Cloud Load Balancing (https://cloud.google.com/load-balancing) features to maintain flexibility while delivering the static IP address required for these needs.

Other times they may require deployments to operate in isolated environments under their full control. Aigens has found it is simple to deliver its solution to meet such requirements through Google Cloud.

"We can actually deploy our solutions to a Google Cloud account that they own," says Peter. "Deployments are so easy. We can have them ready to go in under 10 minutes."

"With BigQuery and Firebase Realtime Database, handling data warehousing and providing a real-time database to all users has been big. Letting shoppers share a cart and order together in real-time are features we've been able to offer that others have not."

—Peter Liu, Co-Founder, Aigens

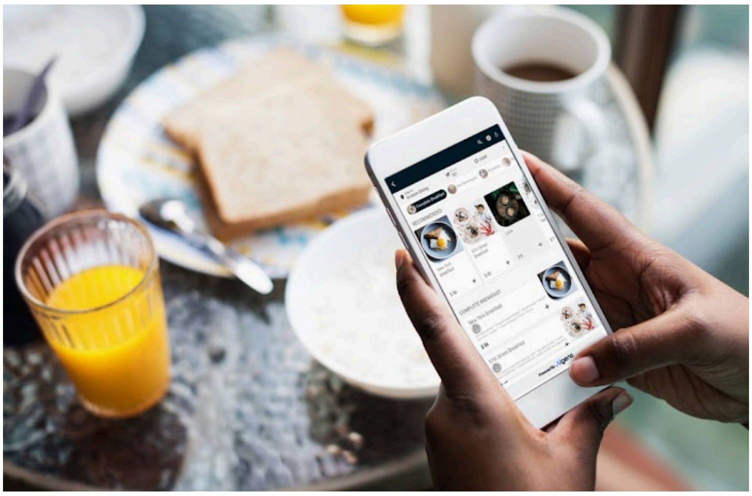
Delivering information in real-time to stores and shoppers

Having adopted Google Cloud early, Peter has seen many new features arrive over the past decade. The biggest stand out in his experience was the arrival of data tools BigQuery (https://cloud.google.com/bigquery) and Firebase Realtime Database (https://firebase.google.com/products/realtime-database/).

"With BigQuery and Firebase Realtime Database, handling data warehousing and providing a real-time database that delivers immediate changes to all users has been big," says Peter. "Delivering real-time status to stores and letting shoppers share a cart and order together in real-time are features we've been able to offer that others have not."

Alongside real-time data performance, Aigens has found the pay-per-use model for Google Cloud data systems has meant they can maintain records that go back to the very early days of the business. Deep data history at scale without prohibitive cost overhead.

From BigQuery, Aigens compiles dashboards and reports to monitor internal KPIs and other useful statistics. "It can process huge amounts of data and return dashboard results in seconds," says Peter.



Efficient and effective DevOps for rapid iteration

Peter holds a clear excitement for testing and exploring new features as they launch on Google Cloud, constantly searching for the next great enhancement for Aigens and its platform.

"I subscribe to news on Google Cloud and whenever there's a new technology I go and try it. Google Cloud is very developer friendly," says Peter. "It lets us get started very quickly and then we can learn more along the way. There's a real ease of use and a feeling that there's a low chance of making some mistakes."

Beyond testing, when new Aigens features are ready for rollout, the team has found it can deploy and update microservices in less than 60 seconds. Canary deployments can be switched to a new version or rolled back in less than 10 seconds.

"We can deploy and migrate traffic seamlessly without any downtime. Being able to do that anytime we want is a big thing for us," says Peter.

Peter also highlights the power of Cloud Logging (https://cloud.google.com/logging) in Google Cloud, centralizing log management in a way that makes it easy to identify insights and issues from the logs with minimum fuss.

Building a marketplace of restaurant digital services

Aigens has been testing capabilities in interesting new areas, including the use of machine learning tools to predict food preparation times or demographic alignment with food recommendations.

But its biggest project for the future is to allow other developers to customize the Aigens platform and become a marketplace for restaurants, hotels, payment platforms, and delivery services to connect through.

"There's new digital platforms and new ways to sell every day," says Peter. "We're trying to build the most complete ecosystem that connects to everybody. Google Cloud provides us with easy to use, innovative technologies that allow us to adopt technology at a fast pace with low costs."

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